

Question number	Answer	Additional guidance	Mark
2(a)	A description that includes four of the following points: • {identify / isolate} gene(s) involved in making beta-carotene in bacteria (1) • cut {DNA sequence / extract (beta carotene) gene} using restriction {endonuclease /enzyme} (1) • insert (beta carotene) gene into vector (1) • method of inserting vector into rice described (1) • grow rice and select for modified rice plants with high yields of beta carotene (1)	ACCEPT obtain gene by reverse transcriptase from mRNA synthesize gene (from a database) ACCEPT gene isolated using restriction enzyme = 2 marks (mp1 and 2) ACCEPT a named vector eg plasmid, liposomes e.g. heat shock, gene gun. micro-injection	(4)

Question number	Answer	Additional guidance	Mark
2(b)(i)	A description that includes three of the following points: • accumulation of genetically engineered material in the environment / other species (1) • cross breeding with other species (1) • unknown long term effects of eating GMOs (1) • allows production of superpests (1)	ACCEPT accumulate in non-target organisms altered gene is found in other species ACCEPT pollen transfer to related species, ACCEPT Named symptom e.g. yellowing of skin, joint pain, dizziness, bleeding, beta carotene is a gene inhibitor, ref allergies credit idea of superweeds/superpests	(3)

Question number	Answer	Additional guidance	Mark
2(b)(ii)	An answer that includes the following points: • appropriate suggestion to reduce risk (1)	accept grow in secure greenhouses, growseparately GM and non GM crops	(1)

